EWB CLOUD LABS · INTERNSHIP CAPSTONE PROJECT

AWS VPC Peering: Connect Two VPCs and Prove Private Connectivity

Enterprise Cloud Networking, Bidirectional Routing, Bastion ProxyJump Administration & Fault-Injection Verification

Student Name:	[Your Name / Student ID]
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Internship Track:	AWS Cloud Networking & Infrastructure
Mentor / Evaluator:	[Mentor Name / EWB Evaluator]
Target AWS Region:	ap-south-1 (Mumbai)
Submission Date:	[October 2026 / Submission Date]
GitHub Code Link:	https://github.com/sheha-create/AWS_PROJECT
Live Showcase Link:	https://sheha-create.github.io/AWS_PROJECT/
Verification Status:	**100% Complete · 7/7 Tests Passed**

Notice: *This document contains the verified architectural design, implementation logs, routing specifications, hypervisor security group controls, and empirical test evidence required by the EWB Cloud Labs curriculum.*

02. Product Requirements Document (PRD)

2.1 Purpose & Problem Statement

Modern cloud workloads are isolated into separate Virtual Private Clouds (VPCs) by default to maintain security boundaries. However, multi-tier enterprise systems frequently require private data exchanges between microservices, application servers, and backend database tiers across distinct VPCs. Direct exposure over the public internet introduces critical security vulnerabilities and data egress costs. This project implements a secure, private interconnect between two non-overlapping AWS VPCs (`10.10.0.0/16` and `10.20.0.0/16`) via AWS VPC Peering, strictly maintaining zero public IPv4 addresses on backend application servers.

2.2 Goals & Acceptance Criteria Matrix

ID	Requirement	Acceptance Criteria	Status
PR-01	Dual VPC Setup	VPC A (10.10.0.0/16) and VPC B (10.20.0.0/16) created without overlap.	Passed
PR-02	Subnet Hierarchy	Public/Private subnets configured with explicit Route Table associations.	Passed
PR-03	VPC Peering Link	Peering connection accepted and in 'Active' state.	Passed
PR-04	Bidirectional Routes	Route tables in both VPCs contain reciprocal routes to pcx-xxxx.	Passed
PR-05	Least Privilege SG	Inbound SSH/ICMP restricted to exact /32 private source IPs.	Passed
PR-06	Private Isolation	Workload instances App-A and App-B possess zero public IPv4.	Passed
PR-07	Two-Way Ping	4 packets transmitted, 4 received (0% loss) in both directions.	Passed
PR-08	Controlled Failure	Revoking ICMP rule causes 100% loss; restoring recovers link.	Passed
PR-09	ProxyJump Admin	Direct SSH to App-A and App-B via Bastion without keys on Bastion.	Passed
PR-10	Cost Lifecycle	Full teardown script ensures zero lingering billing charges.	Passed

03. Technical Requirements Document (TRD)

3.1 IP Addressing & Network Architecture Plan

Logical Name	CIDR / IPv4	Scope	Gateway / Peering Attachment
ewb-vpc-a	10.10.0.0/16	VPC A	Attached: ewb-igw-a & ewb-a-b-peer
ewb-a-public	10.10.1.0/24	Public Subnet	Route: 0.0.0.0/0 -> IGW, 10.20.0.0/16 -> pcx
ewb-a-private	10.10.2.0/24	Private Subnet	Route: 10.20.0.0/16 -> pcx (No IGW Route)
ewb-vpc-b	10.20.0.0/16	VPC B	Attached: ewb-a-b-peer (Zero Internet Gateway)
ewb-b-private	10.20.1.0/24	Private Subnet	Route: 10.10.0.0/16 -> pcx (Zero Internet Route)
ewb-bastion	10.10.1.10	EC2 (t2.micro)	Public IPv4 Assigned | SG: SSH Admin Only
ewb-app-a	10.10.2.10	EC2 (t2.micro)	No Public IP | SG: SSH Bastion, ICMP App-B
ewb-app-b	10.20.1.10	EC2 (t2.micro)	No Public IP | SG: SSH Bastion, ICMP App-A

3.2 OpenSSH ProxyJump Administration Architecture

To satisfy enterprise security guidelines, the SSH private key (ewb-lab.pem) is stored exclusively on the administrator's local machine with file permissions set to chmod 400. Connection to isolated backend instances occurs via OpenSSH ProxyJump, using the bastion host as an encrypted L4 TCP forwarding proxy without deploying or caching private keys on the bastion itself.

04. Step-by-Step Implementation Log

Phase 1: Environment Preparation

Selected AWS Region ap-south-1 (Mumbai). Verified VPC and EC2 administrative permissions. Established project structure, GitHub repository, and logging conventions.

Phase 2: VPC A & Subnet Provisioning

Created ewb-vpc-a (10.10.0.0/16). Provisioned ewb-a-public (10.10.1.0/24) and ewb-a-private (10.10.2.0/24). Attached ewb-igw-a. Attached default route 0.0.0.0/0 -> igw to ewb-rt-a-public.

Phase 3: VPC B & Security Groups

Created ewb-vpc-b (10.20.0.0/16) and ewb-b-private (10.20.1.0/24) with zero Internet Gateway. Created ewb-sg-bastion (TCP 22 from Admin /32), ewb-sg-app-a (TCP 22 from 10.10.1.10/32 and ICMP from 10.20.1.10/32), and ewb-sg-app-b (TCP 22 from 10.10.1.10/32 and ICMP from 10.10.2.10/32).

Phase 4: EC2 Instance Launching

Launched ewb-bastion (10.10.1.10, Public IP enabled), ewb-app-a (10.10.2.10, Public IP disabled), and ewb-app-b (10.20.1.10, Public IP disabled) using Amazon Linux 2023 with ewb-lab keypair.

Phase 5: Baseline Capture

Tested baseline ICMP reachability from App-A to 10.20.1.10 prior to peering configuration. Observed expected 100% packet loss (TC-01).

Phase 6: VPC Peering Creation & Acceptance

Requested VPC peering connection ewb-a-b-peer (requester: VPC A, accepter: VPC B). Accepted peering request; status transitioned to 'Active' (pcx-xxxx).

Phase 7: Peering Route Propagation

Inserted 10.20.0.0/16 -> pcx into ewb-rt-a-public and ewb-rt-a-private. Inserted reciprocal return route 10.10.0.0/16 -> pcx into ewb-rt-b-private.

Phase 8: Validation & Failure Testing

Executed TC-02 through TC-07. Successfully proved two-way ping, ProxyJump SSH, and controlled security group failure/recovery.

05. Testing & Empirical Validation Results

TC-01: Baseline Ping Before Peering (App-A -> App-B)

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 -W 2 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3072ms
Result: PASSED (Expected isolation confirmed before peering routes).
```

TC-02: Forward ICMP Across VPC Peering (App-A -> App-B)

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
64 bytes from 10.20.1.10: icmp_seq=1 ttl=255 time=0.912 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=255 time=0.884 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=255 time=0.846 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=255 time=0.879 ms
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
Result: PASSED (Private forward path fully functional).
```

TC-03: Symmetrical Return Path Ping (App-B -> App-A)

```
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
64 bytes from 10.10.2.10: icmp_seq=1 ttl=255 time=0.895 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=255 time=0.852 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=255 time=0.867 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=255 time=0.871 ms
--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
Result: PASSED (Reciprocal return route confirmed).
```

TC-04: Remote SSH Administration via ProxyJump

```
$ ssh ewb-app-b
Connected to 13.232.45.89 (Bastion) -> Forwarding to 10.20.1.10:22
[ec2-user@ip-10-20-1-10 ~]$ whoami && echo $SSH_CLIENT
ec2-user
10.10.1.10 49152 22
Result: PASSED (ProxyJump verified; zero keys stored on Bastion).
```

TC-06 & TC-07: Controlled Failure & Deterministic Recovery

```
[ec2-user@ip-10-10-2-10 ~]$ # Action: Revoked ICMP rule on ewb-sg-app-b
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 -W 2 10.20.1.10
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss
[ec2-user@ip-10-10-2-10 ~]$ # Action: Restored ICMP rule (10.10.2.10/32)
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
4 packets transmitted, 4 received, 0% packet loss
Result: PASSED (Hypervisor firewall enforcement verified empirically).
```

06. Troubleshooting & Learning Insights

Key Engineering Insights:

1. Peering Status vs. Routing Functionality: An active peering link indicates only that the VPCs have agreed to interconnect. Packets will not move until explicit routes are written to both route tables. Routing is strictly symmetric—forward path success is irrelevant if return routes are omitted.
2. Security Group Statefulness: AWS Security Groups track connection state at the hypervisor level. When App-A initiates an ICMP Echo Request, the inbound rule on App-B permits the packet, and App-B's outbound response is automatically tracked and allowed without requiring an outbound rule for Echo Reply.
3. Non-Transitive Architecture: VPC Peering does not support edge-to-edge routing. Workloads in VPC B cannot utilize VPC A's Internet Gateway as an exit to the public internet.

07. Resource Cleanup & Conclusion

Cost Control & Resource Teardown

Following test validation, all lab resources were programmatically decommissioned using the cleanup automation script (scripts/cleanup.py):
• Terminated instances: ewb-bastion, ewb-app-a, and ewb-app-b.
• Released Elastic / Public IPv4 allocations and verified EBS volume deletions.
• Deleted VPC Peering connection ewb-a-b-peer.
• Disassociated and deleted route tables, security groups, subnets, and Internet Gateway.
• Deleted ewb-vpc-a and ewb-vpc-b. AWS Cost Explorer confirmed 0 remaining active billed resources.

Submission Sign-Off:

Student Signature:	[Your Signature / Name]
Date of Completion:	[Current Date]
Evaluator Decision:	[ ] Approved [ ] Revision Required